

Introduction to Wulver: Resources & HPC

Sep 22, 2026

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Outline

01 What is NJIT Advanced Research Computing?

02 Why and when to use HPC

03 HPC concepts and terminology

04 Getting a login

05 Connecting to Wulver

06 Allocations

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08 Compute partitions

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About NJIT HPC



Partnership

Built through a partnership with DataBank



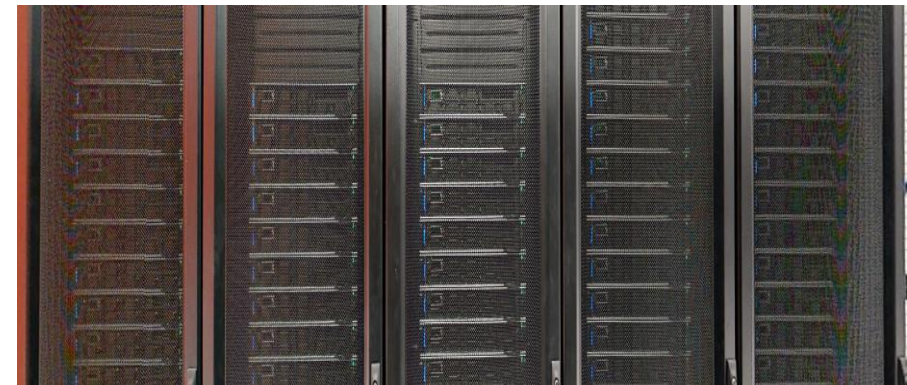
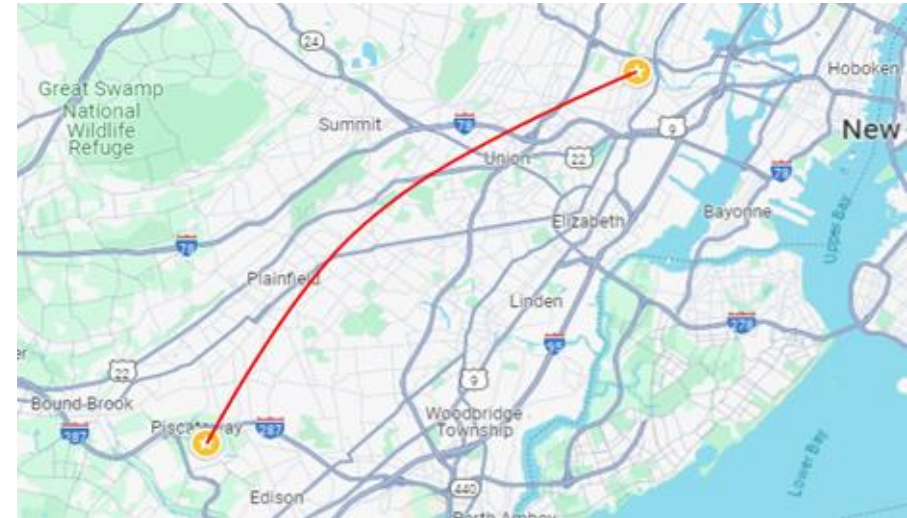
Location

DataBank EWR2 data center, Piscataway, N.J.



Purpose

Supports NJIT research efforts



Service Catalog



Cluster Computing

Built by Dell, the computing environment “Wulver” provides a total of 197 compute nodes or servers



Research Data Storage

High-performance, large capacity data storage spaces that are perfect for a wide variety of research data



Education

HPC and networking resources come together to create an exciting and innovative teaching and research environment



HPC Facilitation Service

Empowering users to perform essential research computing projects through training and effective user support



Scientific Software Development

Deep expertise in developing and deploying software

Why Use HPC?



Your simulations or analyses take too long on your personal computer

- More (faster) cores
- Multithreading
- Multi-node parallelization (openmpi & intelmpi)
- GPU acceleration (NVIDIA's CUDA)
- Distributed computing



The size of your data is too large to be contained (storage) or accessed (memory) on your computer

- Large memory nodes: 512GB; 2 TB
- Distributed computing
- Project storage: TBs range

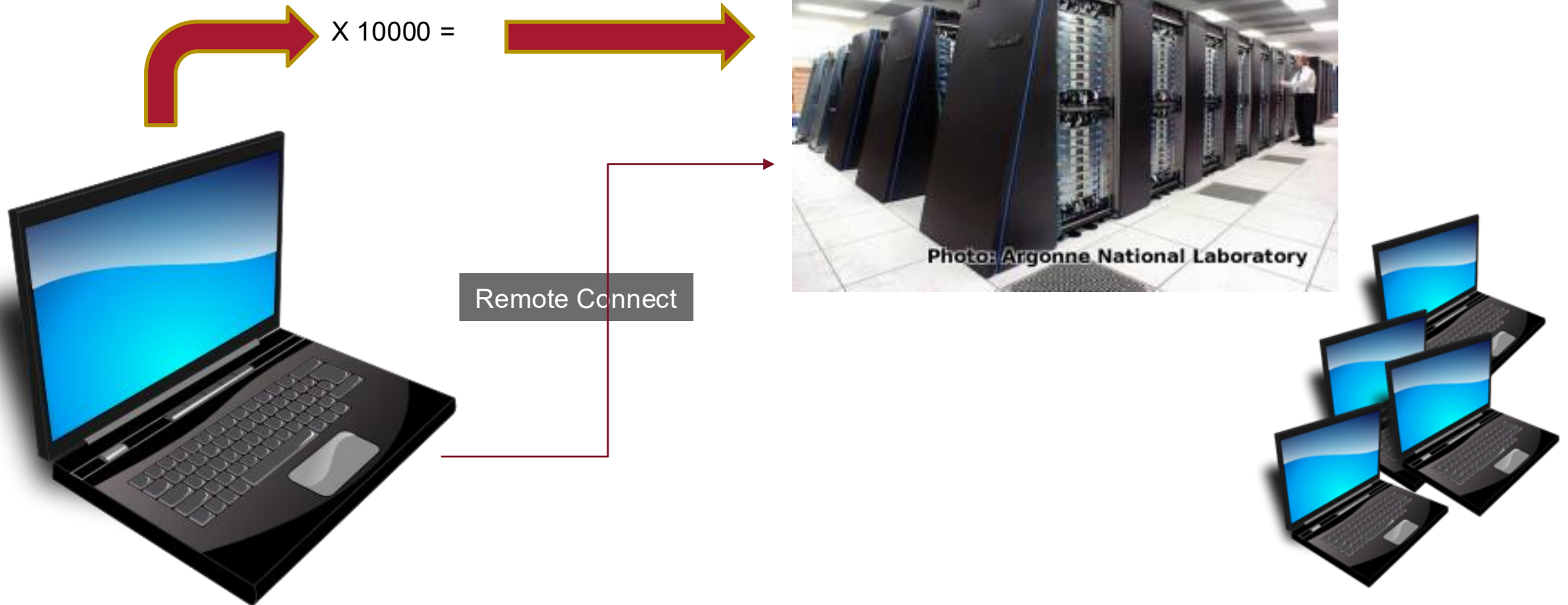


You need a particular software or package for your work

When HPC is not useful

- **Small-scale tasks:** Suitable for desktops or laptops (e.g., basic data analysis, simple simulations).
- **Real-time processing:** Tasks requiring immediate interaction or response (e.g., interactive graphics, live streaming).
- **Lack of parallelism:** You are running a serial code.
- **Running databases**

What is the difference between your laptop and a supercomputer?



HPC Terminology



Compute Node

Equivalent to a high-end workstation, part of a cluster.



Compute Cluster

A group of nodes connected by a high-speed network, forming a supercomputer.



Core

A processing unit inside a CPU; each processor chip holds many cores.



GPU

Graphics processing unit: a separate multi-core processor that handles many small calculations at once.

Memory



Holds data that is being calculated on, as well as computational instructions



Memory types

- *Shared memory* is local to one node and several process
- *Distributed memory* is on multiple nodes



Each core has an associated amount of memory

- Standard nodes: ~4 GB/core
- Huge memory nodes: ~15 GB/core

Getting Access to Wulver

Every account on Wulver is sponsored by NJIT faculty — the request has to come from the sponsor.

Faculty / PI

Email hpc@njit.edu directly. Your own account and your group allocation are set up together.
Email the same address later to add or remove a group member, or to request more storage.

Student / Postdoc

Ask your advisor to email hpc@njit.edu approving your access and naming the group account your jobs are charged to.
Copy your advisor on the first email — leaving them off is the most common delay.

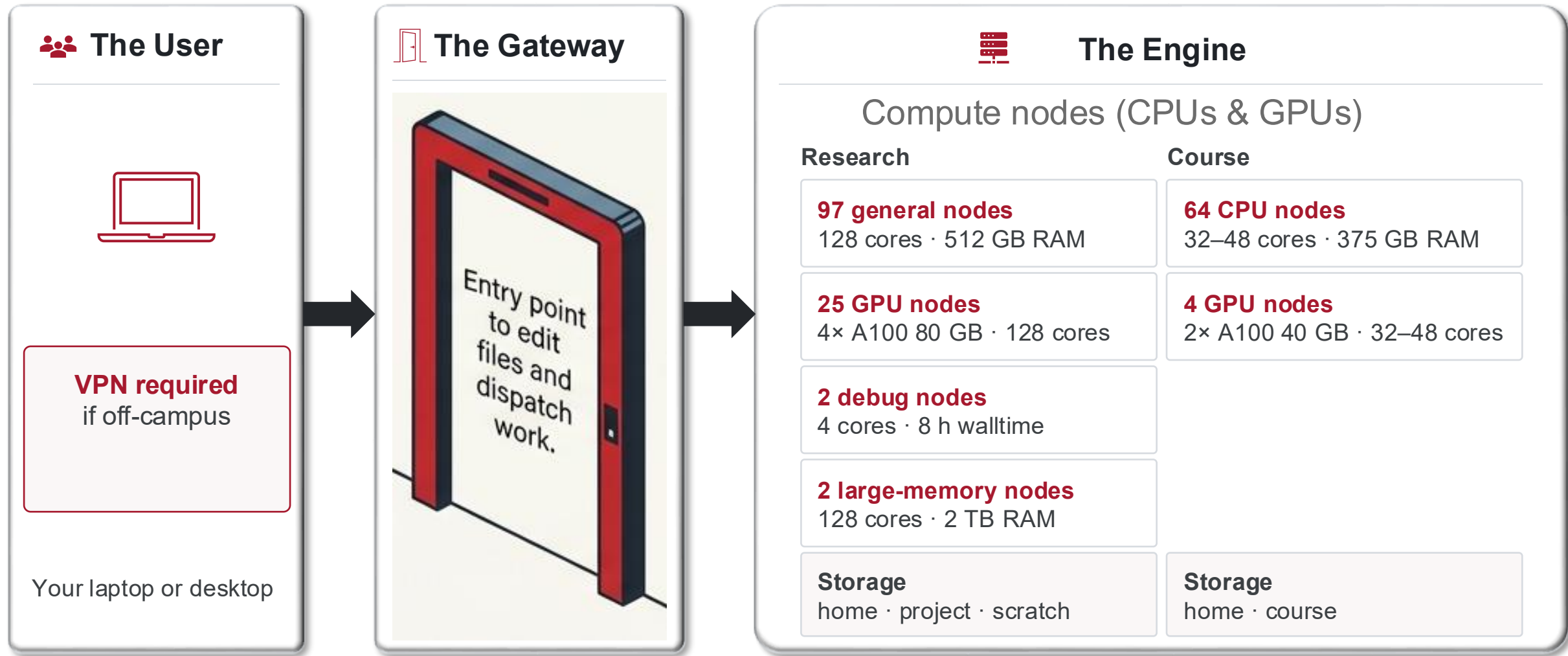
Course

The instructor submits the HPC Course Request Form with the students' UCIDs.
The default is 2,500 Service Units and 100 GB per student, and accounts run through the end of the semester.

Every request includes

- UCID for each person — the short NJIT username such as ab123, not the nine-digit student ID
- The PI or group account the usage should be charged to
- A one-line description of the work the account is for
- Software the user expects to need, so we can confirm what is already in the module stack

How you connect to Wulver



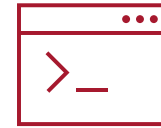
Choosing your gateway

Open OnDemand



- **Interface:** Graphical (GUI), in your browser
- **Best for:** Visual tasks, file transfers, and interactive apps like Jupyter or MATLAB
- **Get there:** ondemand.njit.edu

SSH / Terminal



- **Interface:** Command line (CLI)
- **Best for:** Power users, automated scripts, and fast, lightweight access
- **Get there:** `ssh $UCID@wulver.njit.edu`

Login Nodes – Usage

- Purpose
 - Submit jobs to batch system
 - Edit files
 - Manage your files
 - Interactive work – small scale
- Limits
 - 20GB memory
 - CPU usage is limited to 25% per user
 - Processes are limited to 100 per user
- **Use the batch system for serious computing**



Warning

Do not run VS Code on a login node. It opens dozens of processes and may lock you out of the login node.

Start it from **Open OnDemand** instead, so it runs on a compute node.

HPC Allocations

Every group gets storage and compute; the amounts depend on whether the work is a course or research.

Storage

- Home (~50 GB per user) — limited quota, not for primary storage

Computing time

- 1 SU = CPUs × walltime in hours × usage factor
- Jobs draw on the group account named in the job script

Research

- Project — long-term storage for the group 2 TB
- Scratch — short-term only, purged regularly 10 TB per PI group.

Course

100 GB per student under /course/, readable for one year after the course ends.

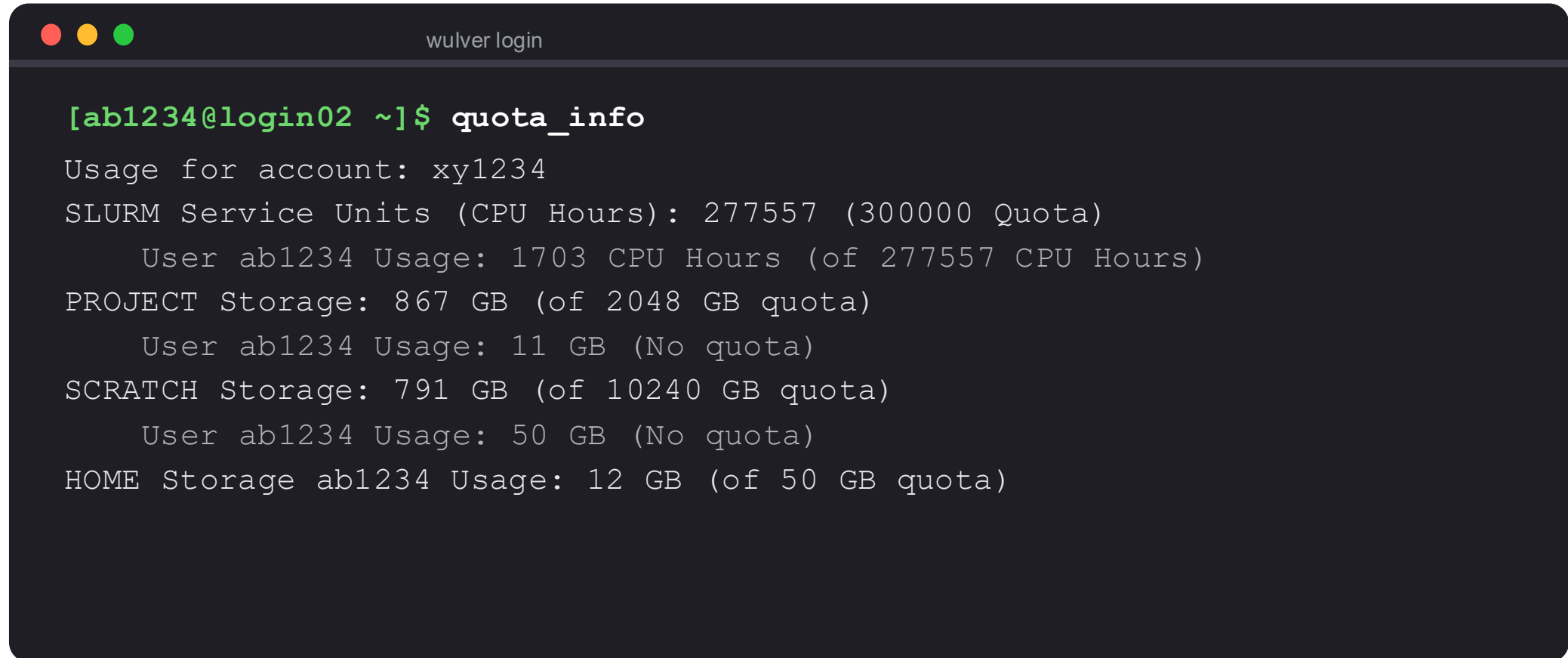
Research

300,000 SUs per PI group per year, the standard annual allocation.

Course

2,500 SUs per student by default, through the end of the semester.

Check Allocations usage



A terminal window titled "wulver login" with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal displays the output of the command `quota_info` for user `ab1234`. The output shows usage for the account `xy1234`, including SLURM Service Units (CPU Hours) and storage usage for PROJECT, SCRATCH, and HOME directories.

```
[ab1234@login02 ~]$ quota_info
Usage for account: xy1234
SLURM Service Units (CPU Hours): 277557 (300000 Quota)
  User ab1234 Usage: 1703 CPU Hours (of 277557 CPU Hours)
PROJECT Storage: 867 GB (of 2048 GB quota)
  User ab1234 Usage: 11 GB (No quota)
SCRATCH Storage: 791 GB (of 10240 GB quota)
  User ab1234 Usage: 50 GB (No quota)
HOME Storage ab1234 Usage: 12 GB (of 50 GB quota)
```

Wulver filesystems

Home \$HOME

Non-research files: profile and shell history.

50 GB

Backed up daily · never purged

Project /project

Active research by the group.
/project/\$PI_UCID/\$LOGIN

2 TB per PI group

More can be purchased

Backed up daily · never purged

Scratch /scratch

Intermediate results, downloads and checkpoints.

/scratch/\$PI_UCID/\$LOGIN

10 TB per PI group

No backup · purged after 30 days

Compute /tmp

Very high-speed temporary space on the node itself.

Varies, about 800 GB

No backup · purged when the job ends

Research /research

Long-term archive. Groups buy as much as they need.

/research/\$PI_UCID

Purchase required

Backed up daily

Course /course

Storage for coursework

/course/<year>/<semester>/
<course>/\$instructor_UCID

100 GB per user

Backed up daily · never purged

Moving data to and from Wulver

rsync upload

```
rsync -avzP /local/path  
$UCID@wulver.njit.edu:/dest
```

Best for large or repeated transfers

rsync download

```
rsync -avzP  
$UCID@wulver.njit.edu:/source  
/local/path
```

Resumes if interrupted

scp one-off copies

```
scp [option] source target
```

Either side can be the remote path

Add -r for a folder

scp example

```
scp -r example  
$UCID@wulver.njit.edu:/home/dir
```

Copies the local example folder into /home/dir

OnDemand Files

ondemand.njit.edu, then Files:
upload and download in the browser.

Small files only

Before you transfer

On campus: NJITsecure. Off campus: NJIT VPN.
UCID password plus Duo two-factor.

Large datasets: use rsync

Partitions

general

Nodes 97 · 128 cores
CPU 2.5 GHz EPYC 7763 (2)
Memory 512 GB

gpu

Nodes 25 · 128 cores
CPU 2.0 GHz EPYC 7713 (2)
GPU 4× NVIDIA A100/Node · 512 GB
MIG: 4 × (2×10G, 20G, 40G) (4 Nodes)

bigmem

Nodes 2 · 128 cores
CPU 2.5 GHz EPYC 7763 (2)
Memory 2 TB, request must be more than 500 GB per job
Jobs 1 per user

debug

Nodes 1 · 4 cores
CPU 2.5 GHz EPYC 7763 (2)
Memory 512 GB

debug_gpu

Nodes 1 · 4 cores
CPU 2.0 GHz EPYC 7713 (2)
GPU MIG: 2×10G, 20G, 40G
Jobs 1 per user
Walltime 8 hours

course

Nodes 64 · 32-48 cores
Memory 375 GB

course_gpu

Nodes 4 · 32-48 cores
MIG 8 A100_10G/Node

Available software on Wulver

```
module load [software_name]
```



CFD

ANSYS *
COMSOL *
OpenFOAM



Chemistry & MD

GROMACS
LAMMPS
VASP *



Programming

GNU / Intel compilers *
MATLAB *
Python



Visualization

ParaView
Tecplot *



Need something else?

Open-source packages can be installed on [request](#).
Have a license? We can help you use it on Wulver.

* licensed software

Reminder

Monthly Maintenance

- Wulver will be temporarily out of service for maintenance once a month, specifically on the 2nd Tuesday, to perform updates, repairs, and upgrades.
- During the maintenance period, the logins will be disabled
- Jobs that do not end before the maintenance window begins will be held until the maintenance is complete
 - Reduce the walltime in the job script to run the job

Office Hours

- Date: Every Monday and Wednesday
- Time: 12:00 - 2:00 pm (Mon), 2:00–4:00 p.m. (Wed)
- Location: GITC 5302M
- Zoom: [Zoom Link](#)
- Meet with our student consultants and ask any questions you have about using HPC resources.
- There's no need to create a ticket in advance.

Wulver is shared, so use it fairly

Login nodes

Never compute on a login node. Send the work to compute nodes with
sbatch srun OnDemand

Your password

Never give it to anyone. Your account is yours alone, and anything run under it is yours.

Job pattern

No floods of very short jobs. Group runs under a minute into one job.

Scratch space

Not long-term storage. Never touch timestamps or shuffle files to dodge the purge.

Warning

Wulver is one system shared by every research group on campus, and working around the rules makes everyone else wait longer.

Gaming the system can have your access revoked, and your PI is notified.

More often the penalty is quieter: your jobs simply run far slower.

On a deadline?

When Wulver is busy your job waits. Two national programs allocate compute to US researchers at no cost:

- **ACCESS** — CPU, GPU, storage and cloud. access-ci.org
- **NAIRR Pilot** — AI compute, datasets and models. nairrpilot.org

Both need an ACCESS ID; ask us which tier fits.

Where to get help

Documentation

- HPC Documentation: <https://hpc.njit.edu/>
- Getting Started: [Access to Wulver](#)
- List of Software: [Wulver Software](#)
- Trainings and Webinars: [HPC Workshops](#)

How-Tos & Software

- [Conda Documentation](#) Installing Python packages via Conda
- MIG: [MIG](#)
- Request Software: [HPC Software Installation](#)
- Access to OnDemand [Open OnDemand](#)

Get Help

- Contact: Please visit [HPC Contact](#)
- Open a ticket: email to hpc@njit.edu
- Consult with Research Computing Facilitator: [Facilitator Calendar Appointment](#)

System updates

- Read Message of the Day on login
- Visit [NJIT HPC News](#)



NJIT

 hpc@njit.edu

 hpc.njit.edu